Stick-together calculations

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Learning target: Find a common final velocity using total momentum.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 5.*
15 kg·m/s right, so the vector sum remains zero.

02. Worked example: explain the model
Both bodies share one final velocity; total mass is 3 kg and total momentum remains 6.

03. Read and interpret the model
Both masses because both share final velocity.

04. Apply the physics
6 kg·m/s; 3 kg; 2 m/s right.

05. Explain your reasoning
When the masses are equal; generally use a mass-weighted average.

06. Review: connect an earlier idea *Review: Unit 2, Day 12.*
No. Tangent velocity arrows change direction, so inward acceleration and net force are needed.

07. Exit check
0 m/s.

08. STRETCH: test the model
Total mass 12/2=6 kg, so stationary cart mass 4 kg.

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01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 5.*
Other velocity -6 m/s; total K=240 J, supplied by released internal energy.

02. Worked example: explain the model
Both bodies share one final velocity; total mass is 3 kg and total momentum remains 6.

03. Read and interpret the model
Both masses because both share final velocity.

04. Apply the physics
p=7; vf 1.4 m/s; initial KE 26.5 J; final 4.9 J; transformed 21.6 J.

05. Explain your reasoning
When the masses are equal; generally use a mass-weighted average.

06. Review: connect an earlier idea *Review: Unit 2, Day 12.*
Both become 1/4; $a_c=v^2/r$ and $F_{net}=mv^2/r$.

07. Exit check
3 m/s; KE 9 J versus 90 J, 10% remains.

08. STRETCH: test the model
First total momentum zero, final v=0. Second total 4 kg·m/s, final v=4/3 m/s right.